Validity Evidence for the Spiritual Harm and Abuse Scale (SHAS)

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Contents

1 About this Project **3**
1.1 Research Questions 3

2 Background of the Project **5**
2.1 Spiritual Harm and Abuse 5
2.2 The Spiritual Harm and Abuse Scale (SHAS) 5
2.3 Validity Research for the SHAS 5

3 Content Validity **6**
3.1 About Content Validity 6
3.2 Content Specification 6
3.3 Summary 8

4 Data Collection **9**

5 Data Preparation **10**
5.1 Load data 10
5.2 Select items 10
5.3 Select item clusters 11
5.4 Dichotomized data 11
5.5 Missing data 12
5.6 Final datasets 14

6 Item Analysis **15**
6.1 Item response theory 15

7 Reliability **40**
7.1 About reliability 40
7.2 Reliability in Classical Test Theory (CTT) 40
7.3 Reliability in Item Response Theory (IRT) 43
7.4 Summary 43

8 Scale Analysis **44**
8.1 Person Fit 44

8.2 Scale Invariance 44
8.3 The Item-Person Map 44
8.4 Information curves 46
8.5 Person Separation Reliability 46
8.6 Item Separation 46
8.7 Summary 46

9 Discussion **47**

References **48**

Chapter 1

About this Project

This is a reproducible research project for validation of the nascent Spiritual Harm and Abuse Scale (SHAS), developed by Dan Koch. It aims to function as an evolving technical report to conceptualize and document various validity studies – and, by extension, validity evidence – for this scale. We hope some of these validity considerations and empirical studies amount to meaningful publishable contributions to the nascent research literature in spiritual abuse and/or the literature in religion, counseling psychology, and/or psychological measurement.

The project is open source, for collaboration. It is built in R using the R Studio IDE [R Core Team, 2022]. No Excel, no SPSS, no Word. It is built in `bookdown` (Xie [2022]) and hosted at Github as a public repository at https://github.com/jackbhuber/SHAS. There it is served over Github Pages as a website – https://jackhuber.org/SHAS/ – for collaboration between Jack and Dan.

1.1 Research Questions

Validation is research into the meanings and uses of scores from instruments. It can be an ongoing research agenda. Following is an initial set of questions to guide research and scale improvement of the SHAS:

- **Item analysis and refinement** - Do some items have too many response categories?
- **Data reduction** - As written and executed, which items appear to be critical (best performing) and which appear to be redundant or noisy? What might a shorter, streamlined instrument look like?
- **Person fit** - How well does the current scale fit the population of examinees? Are there ceiling or floor effects? To what extent might the existing items under-represent the full range of abuse?

- **Scale invariance** - Do the scales mean the same thing to different groups of examinees? To what extent do these scales (abuse constructs) vary by age, gender, and/or sexuality?
- **Developmental pathway** - Beyond the current conceptualization of scales composed of redundant items, what might an incremental scale of abuse with diagnostic items look like?

Chapter 2

Background of the Project

2.1 Spiritual Harm and Abuse

This is where Dan can introduce and discuss spiritual harm and abuse – his history with the topic, what it might mean in real world terms, the need for scientific study of the topic, and research to date. At a minimum he can paste in text from his extant writings and/or write new text.

2.2 The Spiritual Harm and Abuse Scale (SHAS)

This might be where Dan presents the case for a clinical measure of spiritual harm and abuse and the work he has done to date to develop the SHAS. This could be a summary of his forthcoming JSSR article.

2.3 Validity Research for the SHAS

This might be where I discuss what it means to conduct validity research on any instrument, and in particular the SHAS.

Chapter 3

Content Validity

In this chapter, we consider the content validity of the Spiritual Harm and Abuse Scale (SHAS). We begin by defining content validity and discussing its importance. Then we document the work we did to map the domain of spiritual abuse for the purpose of developing items for piloting and operational use. We conclude by considering what scope exists for more work in this area that would improve future development of SHAS instrumentation.

3.1 About Content Validity

The focus of content validity is "the relationship between the content of a test and the construct it is intended to measure" [American Educational Research Association et al., 2014]. Do they match? Does the instrument contain content that is not part of the construct? Are there parts of the content unmeasured, such that a high score still underrepresents an examinee's true level of a trait?

Content validity in this case concerns the relationship between the concept of spiritual abuse and the content of the 66 items that were piloted to form the SHAS. What is spiritual abuse? What are its important parts? What counts as evidence of it? Is there more to spiritual abuse than was captured by the 66 items?

3.2 Content Specification

"Test developers often work from a specification of the content domain. The content specification carefully describes the content in detail, often with a classification of areas of content and types of items" [American Educational Research Association et al., 2014]. For the purpose of piloting items, the domain of spiritual abuse included two constructs: external (abusive) events and internal states (of emotional harm).

3.2.1 External Events

One domain of spiritual abuse is external abuse *events*. The focus is on something that happened, or a pattern of recurring behavior on the part of someone in a place of spiritual authority that was experienced by the examinee as abusive. The first 52 items of the SHAS were drawn from this domain of possible abusive events. To each item, an examinee was offered five response categories:

1. Never
2. Once or twice
3. Sometimes
4. Often
5. Always

To sum each examinee's vector of responses to these items is to accumulate evidence of abuse. In the aggregate, scores for this collection of items could range from 52 to 260. An examinee who *never* experienced *any* of these events would score 52. At the other extreme, an examinee who scored 260 had *always* experienced *all* of these events.

3.2.2 Internal States

Another important domain of spiritual abuse is *internal states*. The focus is on the subjective experience of an examinee "as a result of negative experiences in church or Christian group settings." The questionnaire included 14 items asking examinees about their emotions in response to events. To each item, an examinee was offered the same five response categories as before:

1. Never
2. Once or twice
3. Sometimes
4. Often
5. Always

To sum each examinee's vector of responses to these items is to accumulate evidence of internal emotional *harm*. Total scores for these items could range from 14 to 70. Examinees who *never* experienced *any* of these unpleasant emotions would score 14, while examinees who scored 70 had *always* experienced *all* of the difficult emotions.

3.2.3 Additional Domains

Reise and Waller [2009] make some interesting points about the nature of the traits addressed in clinical settings which provoke thought about the trait of spiritual abuse.

3.3 Summary

What counts as spiritual abuse? What counts as evidence of it?

Chapter 4

Data Collection

The data were collected online from January 28, 2021 to February 27, 2021.

A final total of 3,222 people responded to the survey at some level. Depending on how much data are missing, this is an impressively large sample size which will provide lots of power for most simple statistics and should for the most part satisfy the large-sample requirements of the advanced statistical procedures associated with measurement work (i.e., factor analysis, IRT).

However, the data come from a convenience sample which raises some challenge of sampling bias. Dan sent the link out to [lots of people], and (if memory serves) encouraged them to send it to people *they* knew. This would seem to bias responses toward people demographically similar to Dan: white, Millennial, west coast American, middle class, highly educated, primarily Protestant and Catholic Christian, etc. This will be more apparent upon closer inspection of the demographics of the sample.

One source of examinees was people who listen to progressive Christians podcasts such as *You Have Permission* which Dan produced at the time of data collection. This is a known source of bias toward people who have some interest in Christianity along with at least some level of rejection of conservative/orthodox Christianity and/or political conservatism. However, this would be an appropriate group from which to invite participation if the target population is people who have experienced spiritual abuse.

Chapter 5

Data Preparation

This chapter documents our decisions surrounding how to prepare the data for analysis. Here we get into the weeds. We begin with loading the data from the original data file. From this we select a subset of just the items to which we apply labels and recode so that they begin from 0. We deal with missing data. The chapter concludes with a final set of datasets for use in the remaining chapters.

5.1 Load data

The data are contained in an Excel file, simply named "data.xlsx", in the "data" folder of this project. The important thing is to keep all the data in the same folder within the larger "SHAS Validity" project folder.

```
# LOAD ORIGINAL DATA
library(readxl)  # for reading the source Excel file
shas <- read_excel("data/data.xlsx")  # load Excel file
```

5.2 Select items

Which items should we include for analysis? The first 13 columns in the original data file report data from several screening items that were placed at the beginning of the questionnaire. But because they are constant and not variable, they don't seem to have a place in scale development. Therefore it seems the important columns range from columns 14 to 97. The first set of items measure the **External Events** scale. These 52 items range from columns 14 to 35. The second set of items measure the **Internal States** scale, and these 14 items range from columns 66 to 79. The last set of items capture **demographic** variables, and these range from columns 80 to 97.

```r
# SELECT ITEMS
all <- shas[, 14:97]  # select items, dems from columns 14-79
all <- subset(all, select = -c(Ext_Avg:Global_Avg))  # remove columns for now
items <- all[, 1:66]
```

5.3 Select item clusters

We select two subsets of the item data, one from just the External Events items and the other from the Internal States items.

```r
ext_items <- items[, 1:52]
int_items <- items[, 52:66]
```

5.4 Dichotomized data

Some IRT models (such as the Rasch model) and their estimation procedures in various R packages are based on dichotomous data (0=incorrect, 1=correct) rather than the polytomous data that comes from Likert scales. It is also possible that the extra categories create noise and improved SHAS items might use fewer response categories. Here we create several data sets of dichotomized SHAS items.

```r
# DICHOTOMIZED DATA SETS
ditems <- items
ditems[ditems == 1] <- 0
ditems[ditems == 2] <- 1
ditems[ditems == 3] <- 1
ditems[ditems == 4] <- 1
ditems[ditems == 5] <- 1

dext_items <- ext_items
dext_items[dext_items == 1] <- 0
dext_items[dext_items == 2] <- 1
dext_items[dext_items == 3] <- 1
dext_items[dext_items == 4] <- 1
dext_items[dext_items == 5] <- 1

dint_items <- int_items
dint_items[dint_items == 1] <- 0
dint_items[dint_items == 2] <- 1
dint_items[dint_items == 3] <- 1
dint_items[dint_items == 4] <- 1
dint_items[dint_items == 5] <- 1
```

5.5 Missing data

Missing data can be an important problem. The problem has two parts. The first is how many data are missing. Obviously the more data are missing, the larger the barrier this problem poses to analysis. The second is the mechanism of missing data. Are the missing data randomly, or is there a pattern to the missing data - like a problematic item, or a distinct group of examinees who chose not to respond? Figure 5.1 presents two plots from the VIM package for missing data [Templ et al., 2021].

```
# VISUALIZE AND QUANTIFY MISSING DATA
library(VIM)
a <- VIM::aggr(all, plot = FALSE)
plot(a, numbers = TRUE, prop = FALSE)
```

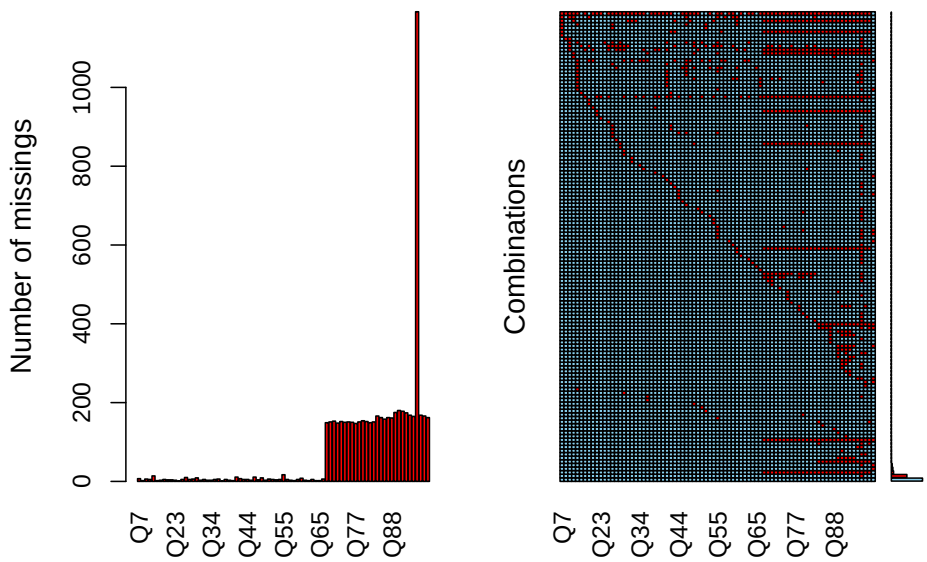

Figure 5.1: Missing Data

The first plot in Figure 5.1 is a frequency distribution of the number of missing data entries for each item. This plot reveals very few missing data for the first set of External Events items, but nearly 200 missing entries for the second set of Internal States items and demographics. The raises two questions. The first is: Who were those examinees? We know that they responded to the first 52 items, then apparently quit the survey. Why? Was it just fatigue, or is there more to know about them? The second question is what to do with the information they actually did provide.

The second plot in Figure 5.1 illustrates the different combinations of missing entries. The blue squares represent cases with complete responses, and the red indicates missing data. The rows represent combinations of complete and

missing responses. The bottom row of entirely blue squares represents complete data – examinees who responded to all of the items. The barplot on the right axis indicates that these were the most frequent combinations. The second most frequent combination was examinees who completed all of the External Events items and then quit the survey. Beyond that, there is no apparent order or pattern to the missing data.

Mechanism of missing data. McKnight et al. [2007] prescribe several approaches for investigating the mechanisms of missing data. An appropriate first step is to test whether the data are Missing Completely At Random (MCAR) with the use of Little's test Little [1988]. This is a chi-square test of the difference between the observed pattern of missing responses and the pattern of responses that would expected if missingness were completely random. We report the results of this test in Table **??**.

```
library(naniar)
library(knitr)
knitr::kable(mcar_test(all), digits = 2)
```

statistic	df	p.value	missing.patterns
2089.34	8663	1	117

The test is fairly conclusive. The p-value for the chi-square is 1, which means observed pattern of missing data do not diverge significantly from what would be expected if the data were missing completely at random.

Missing data methods. There are several different ways to handle missing data. A common one is to *delete all cases missing any data*. This is called **listwise** deletion of data. Its primary advantages are simplicity, expedience, and analysis based on complete information. Its disadvantages are (1) disposal of the information offered by those examinees who responded to some items but not others, and (2) diminishment of the dataset, and with it, a reduction in statistical power. There are also methods of imputing values to missing responses. Here, for expedience, we use listwise deletion to create several datasets with complete data.

```
# LISTWISE DELETION OF MISSING DATA
all_l <- na.omit(all)  #  All items + demographics
items_l <- na.omit(items)  # All Likert items
ext_items_l <- na.omit(ext_items)  # Likert items for External Events post listwise
int_items_l <- na.omit(int_items)  # Likert items for Internal States post listwise
ditems_l <- na.omit(ditems)  # All dichotomized Likert items post listwise
dext_items_l <- na.omit(dext_items)  # Dichotomized External Events items post listwise
dint_items_l <- na.omit(dint_items)  # Dichotomized Internal States items post listwise
```

5.6 Final datasets

```
# SAVE DATA FILES
write.csv(all, "data/all.csv", row.names = FALSE)
write.csv(items, "data/items.csv", row.names = FALSE)
write.csv(ext_items, "data/ext_items.csv", row.names = FALSE)
write.csv(int_items, "data/int_items.csv", row.names = FALSE)
write.csv(all_l, "data/all_l.csv", row.names = FALSE)
write.csv(items_l, "data/items_l.csv", row.names = FALSE)
write.csv(ext_items_l, "data/ext_items_l.csv", row.names = FALSE)
write.csv(int_items_l, "data/int_items_l.csv", row.names = FALSE)
write.csv(ditems_l, "data/ditems_l.csv", row.names = FALSE)
write.csv(dext_items_l, "data/dext_items_l.csv", row.names = FALSE)
write.csv(dint_items_l, "data/dint_items_l.csv", row.names = FALSE)
rm(list = ls())
```

Chapter 6

Item Analysis

In this chapter, we examine the technical quality of the items that Dan piloted for the SHAS. Here we address questions like the following:

- Which items, if any, are too "easy"? What is the impact of this?
- Which items, if any, are too "difficult"? What is the impact of this?
- Which items, if any, correlate weakly with others (and therefore might need to be refined or excluded from scaling?
- Which items, if any, have too many response categories which might be profitably collapsed?

We begin by defining item analysis and discussing its importance. Then we document our item analyses from an item response theory (IRT) framework. We conclude by summarizing what we learn from these analyses to help us refine these 66 items and possibly develop better ones in the future.

6.1 Item response theory

Item response theory (IRT) is an established framework for empirically evaluating what items measure [Embretson and Reise, 2000, Wright and Masters, 1982]. IRT methods complement methods derived from classical test theory in several ways.

1. **IRT places both examinees and items on the same interval scale**. In classical test theory, examinees differ in their scores and items differ in their means or percent of examinees selecting a response; examinees and items are on different scales. By contrast, IRT methods allow measurement of persons and items on the ***same scale*** with *equal interval properties* of the *logit* scale** and resulting linear measures (Wright & Stone, 1999). In IRT we can estimate *item parameters independently of the characteristics*

of the calibrating sample, and we can estimate *person parameters apart from the difficulties of the items taken* (Masters, 1982; Rasch, 1966).

2. **Parameters for examinees and items derived from IRT are independent of each other**.

3. **IRT gives items diagnostic value**. A core assumption of CTT is that items are *redundant*; they contribute meaningful variation to a total score but they do not occupy a distinct place on the score continuum. IRT is different. IRT does not assume that items are redundant; instead it investigates whether items make distinct contributions to the measurement of the latent trait. Whereas classical test theory measurement is principally concerned with the ratio of noise to signal in the variation contributed by an item, IRT models compute the probability of a certain response to an item given the level of the latent construct the individual possesses (i.e., spiritual abuse) and the relevant item's difficulty of endorsement. IRT is based on the logistic model and operates on the *logit* scale.

In what follows, we conduct item analyses through a comparison of several different IRT models: the Rating Scale Model (RSM), the Partial Credit Model (PCM), the Generalized Partial Credit Model (GPCM).

6.1.1 IRT Model Assumptions

IRT makes two assumptions of the data. The first is unidimensionality of the latent trait. Is the trait one dimension? This is important for the SHAS because the questionnaire explicitly distinguished between items measuring external events and internal states. The second assumption is local independence [Embretson and Reise, 2000]. This means that, once the latent variable has been accounted for, any set of items should not share any meaningful correlation (Edwards, Houts, & Cai, 2018).

6.1.1.1 Unidimensionality

To verify the assumption of unidimensionality we conducted three analyses. First we confirmed the factor structure using confirmatory factor analysis (CFA). Then we determined the proportion of variance in the observed item responses due to the Rasch model. Finally, we conducted Rasch Principal Component Analysis of Residuals (PCAR) (Smith, 2002).

Factor structure.

Proportion of variance.

Reckase suggests that the unidimensionality assumption is safely met if the Rasch model explains 20% of the variance in the data. According to this analysis, the Rasch model explains 44% of the variance in the SHAS item data. Thus we have some evidence of unidimensionality with which to proceed to further Rasch modeling.

Principal Component Analysis of Residuals (PCAR)

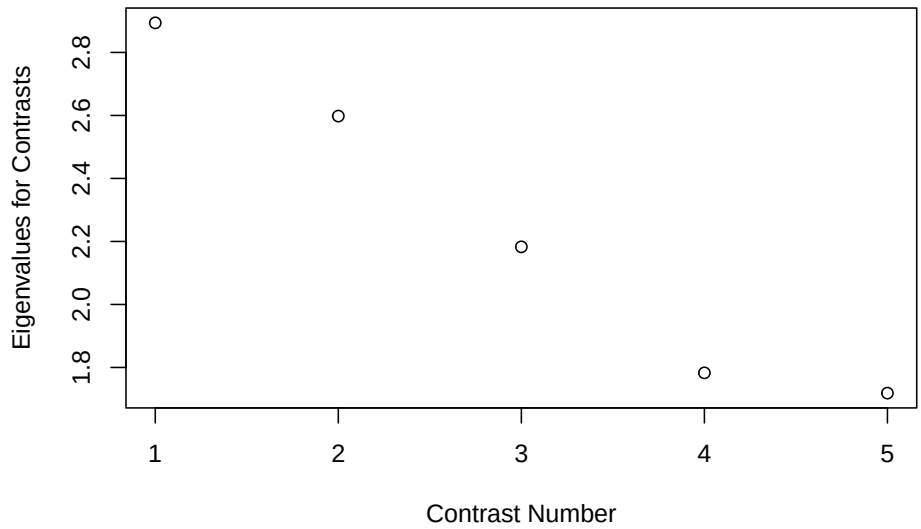

Figure 6.1: Contrasts from PCA of Standardized Residual Correlations

The second analysis of PCAR examines the correlations among residuals. The premise is that once the Rasch model has been estimated, the correlations among the residuals should be minimal. Linacre () suggests that contrasts with eigenvalues of 2.0 or below can be considered noise. In Figure 6.1, our PCAR analysis found three contrasts that rose above that 2.0 threshold.

6.1.1.2 Local Independence

To check the assumption of local independence, we used the Q3 statistic (Yen, 1984). The Q3 statistic index criteria are that the raw residual correlation between pairs of items should never exceed 0.10 (Marais & Andrich, 2008).

```
------------------------------------------------------------
Semiparametric Marginal Maximum Likelihood Estimation
Raschtype Model with generalized logistic link function: alpha1= 0 , alpha2= 0
------------------------------------------------------------
..........................................................
Iteration 1     2022-05-01 11:52:50
   Deviance=178730.6812
    Maximum b parameter change=0.837684
..........................................................
Iteration 2     2022-05-01 11:52:51
   Deviance=174132.6014 | Deviance change=4598.079793
    Maximum b parameter change=0.263887
..........................................................
```

```
Iteration 3     2022-05-01 11:52:51
   Deviance=173692.1918 | Deviance change=440.409632
    Maximum b parameter change=0.094586
.............................................................
Iteration 4     2022-05-01 11:52:51
   Deviance=173647.6904 | Deviance change=44.501431
    Maximum b parameter change=0.027496
.............................................................
Iteration 5     2022-05-01 11:52:51
   Deviance=173642.3097 | Deviance change=5.3807
    Maximum b parameter change=0.009127
.............................................................
Iteration 6     2022-05-01 11:52:51
   Deviance=173640.8591 | Deviance change=1.450531
    Maximum b parameter change=0.005552
.............................................................
Iteration 7     2022-05-01 11:52:51
   Deviance=173639.9026 | Deviance change=0.956595
    Maximum b parameter change=0.004379
.............................................................
Iteration 8     2022-05-01 11:52:51
   Deviance=173639.0673 | Deviance change=0.835259
    Maximum b parameter change=0.003935
.............................................................
Iteration 9     2022-05-01 11:52:51
   Deviance=173638.2993 | Deviance change=0.767991
    Maximum b parameter change=0.003732
.............................................................
Iteration 10     2022-05-01 11:52:51
   Deviance=173637.5854 | Deviance change=0.713935
    Maximum b parameter change=0.003601
.............................................................
Iteration 11     2022-05-01 11:52:51
   Deviance=173636.9197 | Deviance change=0.665655
    Maximum b parameter change=0.00348
.............................................................
Iteration 12     2022-05-01 11:52:51
   Deviance=173636.2985 | Deviance change=0.621216
    Maximum b parameter change=0.003363
.............................................................
Iteration 13     2022-05-01 11:52:51
   Deviance=173635.7186 | Deviance change=0.579931
    Maximum b parameter change=0.00325
.............................................................
Iteration 14     2022-05-01 11:52:51
   Deviance=173635.1771 | Deviance change=0.541464
```

```
    Maximum b parameter change=0.00314
.............................................................
Iteration 15     2022-05-01 11:52:51
   Deviance=173634.6715 | Deviance change=0.505589
    Maximum b parameter change=0.003034
.............................................................
Iteration 16     2022-05-01 11:52:51
   Deviance=173634.1994 | Deviance change=0.47212
    Maximum b parameter change=0.002932
.............................................................
Iteration 17     2022-05-01 11:52:51
   Deviance=173633.7585 | Deviance change=0.440891
    Maximum b parameter change=0.002834
.............................................................
Iteration 18     2022-05-01 11:52:52
   Deviance=173633.3467 | Deviance change=0.411751
    Maximum b parameter change=0.002738
.............................................................
Iteration 19     2022-05-01 11:52:52
   Deviance=173632.9622 | Deviance change=0.384558
    Maximum b parameter change=0.002646
.............................................................
Iteration 20     2022-05-01 11:52:52
   Deviance=173632.603 | Deviance change=0.359181
    Maximum b parameter change=0.002558
.............................................................
Iteration 21     2022-05-01 11:52:52
   Deviance=173632.2675 | Deviance change=0.335497
    Maximum b parameter change=0.002472
.............................................................
Iteration 22     2022-05-01 11:52:52
   Deviance=173631.9541 | Deviance change=0.313392
    Maximum b parameter change=0.002389
.............................................................
Iteration 23     2022-05-01 11:52:52
   Deviance=173631.6614 | Deviance change=0.29276
    Maximum b parameter change=0.002309
.............................................................
Iteration 24     2022-05-01 11:52:52
   Deviance=173631.3879 | Deviance change=0.273502
    Maximum b parameter change=0.002232
.............................................................
Iteration 25     2022-05-01 11:52:52
   Deviance=173631.1323 | Deviance change=0.255524
    Maximum b parameter change=0.002158
.............................................................
```

```
Iteration 26     2022-05-01 11:52:52
   Deviance=173630.8936 | Deviance change=0.238741
    Maximum b parameter change=0.002086
.............................................................
Iteration 27     2022-05-01 11:52:52
   Deviance=173630.6705 | Deviance change=0.223073
    Maximum b parameter change=0.002016
.............................................................
Iteration 28     2022-05-01 11:52:52
   Deviance=173630.4621 | Deviance change=0.208444
    Maximum b parameter change=0.001949
.............................................................
Iteration 29     2022-05-01 11:52:52
   Deviance=173630.2673 | Deviance change=0.194784
    Maximum b parameter change=0.001884
.............................................................
Iteration 30     2022-05-01 11:52:52
   Deviance=173630.0853 | Deviance change=0.182029
    Maximum b parameter change=0.001822
.............................................................
Iteration 31     2022-05-01 11:52:52
   Deviance=173629.9151 | Deviance change=0.170117
    Maximum b parameter change=0.001761
.............................................................
Iteration 32     2022-05-01 11:52:52
   Deviance=173629.7562 | Deviance change=0.158993
    Maximum b parameter change=0.001703
.............................................................
Iteration 33     2022-05-01 11:52:52
   Deviance=173629.6076 | Deviance change=0.148604
    Maximum b parameter change=0.001646
.............................................................
Iteration 34     2022-05-01 11:52:52
   Deviance=173629.4687 | Deviance change=0.138899
    Maximum b parameter change=0.001592
.............................................................
Iteration 35     2022-05-01 11:52:52
   Deviance=173629.3388 | Deviance change=0.129835
    Maximum b parameter change=0.001539
.............................................................
Iteration 36     2022-05-01 11:52:53
   Deviance=173629.2174 | Deviance change=0.121368
    Maximum b parameter change=0.001488
.............................................................
Iteration 37     2022-05-01 11:52:53
   Deviance=173629.104 | Deviance change=0.113458
```

```
    Maximum b parameter change=0.001439
.............................................................
Iteration 38     2022-05-01 11:52:53
   Deviance=173628.9979 | Deviance change=0.106068
    Maximum b parameter change=0.001391
.............................................................
Iteration 39     2022-05-01 11:52:53
   Deviance=173628.8988 | Deviance change=0.099163
    Maximum b parameter change=0.001345
.............................................................
Iteration 40     2022-05-01 11:52:53
   Deviance=173628.806 | Deviance change=0.092712
    Maximum b parameter change=0.001301
.............................................................
Iteration 41     2022-05-01 11:52:53
   Deviance=173628.7194 | Deviance change=0.086683
    Maximum b parameter change=0.001258
.............................................................
Iteration 42     2022-05-01 11:52:53
   Deviance=173628.6383 | Deviance change=0.08105
    Maximum b parameter change=0.001216
.............................................................
Iteration 43     2022-05-01 11:52:53
   Deviance=173628.5625 | Deviance change=0.075786
    Maximum b parameter change=0.001176
.............................................................
Iteration 44     2022-05-01 11:52:53
   Deviance=173628.4917 | Deviance change=0.070867
    Maximum b parameter change=0.001137
.............................................................
Iteration 45     2022-05-01 11:52:53
   Deviance=173628.4254 | Deviance change=0.066269
    Maximum b parameter change=0.0011
.............................................................
Iteration 46     2022-05-01 11:52:53
   Deviance=173628.3634 | Deviance change=0.061971
    Maximum b parameter change=0.001064
.............................................................
Iteration 47     2022-05-01 11:52:53
   Deviance=173628.3055 | Deviance change=0.057954
    Maximum b parameter change=0.001029
.............................................................
Iteration 48     2022-05-01 11:52:53
   Deviance=173628.2513 | Deviance change=0.054199
    Maximum b parameter change=0.000995
.............................................................
```

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Iteration 49     2022-05-01 11:52:53
   Deviance=173628.2006 | Deviance change=0.050689
    Maximum b parameter change=0.000962
.............................................................
Iteration 50     2022-05-01 11:52:53
   Deviance=173628.1532 | Deviance change=0.047408
    Maximum b parameter change=0.000931
.............................................................
Iteration 51     2022-05-01 11:52:53
   Deviance=173628.1088 | Deviance change=0.044341
    Maximum b parameter change=9e-04
.............................................................
Iteration 52     2022-05-01 11:52:53
   Deviance=173628.0674 | Deviance change=0.041473
    Maximum b parameter change=0.000871
.............................................................
Iteration 53     2022-05-01 11:52:53
   Deviance=173628.0286 | Deviance change=0.038792
    Maximum b parameter change=0.000842
.............................................................
Iteration 54     2022-05-01 11:52:53
   Deviance=173627.9923 | Deviance change=0.036285
    Maximum b parameter change=0.000814
.............................................................
Iteration 55     2022-05-01 11:52:53
   Deviance=173627.9583 | Deviance change=0.03394
    Maximum b parameter change=0.000788
.............................................................
Iteration 56     2022-05-01 11:52:53
   Deviance=173627.9266 | Deviance change=0.031749
    Maximum b parameter change=0.000762
.............................................................
Iteration 57     2022-05-01 11:52:53
   Deviance=173627.8969 | Deviance change=0.029699
    Maximum b parameter change=0.000737
.............................................................
Iteration 58     2022-05-01 11:52:53
   Deviance=173627.8691 | Deviance change=0.027782
    Maximum b parameter change=0.000713
.............................................................
Iteration 59     2022-05-01 11:52:53
   Deviance=173627.8431 | Deviance change=0.02599
    Maximum b parameter change=0.000689
.............................................................
Iteration 60     2022-05-01 11:52:53
   Deviance=173627.8188 | Deviance change=0.024314
```

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    Maximum b parameter change=0.000667
.............................................................
Iteration 61     2022-05-01 11:52:54
   Deviance=173627.7961 | Deviance change=0.022746
    Maximum b parameter change=0.000645
.............................................................
Iteration 62     2022-05-01 11:52:54
   Deviance=173627.7748 | Deviance change=0.02128
    Maximum b parameter change=0.000624
.............................................................
Iteration 63     2022-05-01 11:52:54
   Deviance=173627.7549 | Deviance change=0.019909
    Maximum b parameter change=0.000604
.............................................................
Iteration 64     2022-05-01 11:52:54
   Deviance=173627.7362 | Deviance change=0.018626
    Maximum b parameter change=0.000584
.............................................................
Iteration 65     2022-05-01 11:52:54
   Deviance=173627.7188 | Deviance change=0.017427
    Maximum b parameter change=0.000565
.............................................................
Iteration 66     2022-05-01 11:52:54
   Deviance=173627.7025 | Deviance change=0.016305
    Maximum b parameter change=0.000546
.............................................................
Iteration 67     2022-05-01 11:52:54
   Deviance=173627.6873 | Deviance change=0.015255
    Maximum b parameter change=0.000528
.............................................................
Iteration 68     2022-05-01 11:52:54
   Deviance=173627.673 | Deviance change=0.014273
    Maximum b parameter change=0.000511
.............................................................
Iteration 69     2022-05-01 11:52:54
   Deviance=173627.6596 | Deviance change=0.013355
    Maximum b parameter change=0.000494
.............................................................
Iteration 70     2022-05-01 11:52:54
   Deviance=173627.6471 | Deviance change=0.012496
    Maximum b parameter change=0.000478
.............................................................
Iteration 71     2022-05-01 11:52:54
   Deviance=173627.6354 | Deviance change=0.011692
    Maximum b parameter change=0.000463
.............................................................
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Iteration 72     2022-05-01 11:52:54
   Deviance=173627.6245 | Deviance change=0.01094
    Maximum b parameter change=0.000448
.............................................................
Iteration 73     2022-05-01 11:52:54
   Deviance=173627.6143 | Deviance change=0.010237
    Maximum b parameter change=0.000433
.............................................................
Iteration 74     2022-05-01 11:52:54
   Deviance=173627.6047 | Deviance change=0.009579
    Maximum b parameter change=0.000419
.............................................................
Iteration 75     2022-05-01 11:52:54
   Deviance=173627.5957 | Deviance change=0.008963
    Maximum b parameter change=0.000405
.............................................................
Iteration 76     2022-05-01 11:52:54
   Deviance=173627.5873 | Deviance change=0.008387
    Maximum b parameter change=0.000392
.............................................................
Iteration 77     2022-05-01 11:52:54
   Deviance=173627.5795 | Deviance change=0.007848
    Maximum b parameter change=0.000379
.............................................................
Iteration 78     2022-05-01 11:52:54
   Deviance=173627.5721 | Deviance change=0.007344
    Maximum b parameter change=0.000367
.............................................................
Iteration 79     2022-05-01 11:52:54
   Deviance=173627.5653 | Deviance change=0.006872
    Maximum b parameter change=0.000355
.............................................................
Iteration 80     2022-05-01 11:52:54
   Deviance=173627.5588 | Deviance change=0.006431
    Maximum b parameter change=0.000343
.............................................................
Iteration 81     2022-05-01 11:52:54
   Deviance=173627.5528 | Deviance change=0.006018
    Maximum b parameter change=0.000332
.............................................................
Iteration 82     2022-05-01 11:52:54
   Deviance=173627.5472 | Deviance change=0.005632
    Maximum b parameter change=0.000321
.............................................................
Iteration 83     2022-05-01 11:52:54
   Deviance=173627.5419 | Deviance change=0.00527
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    Maximum b parameter change=0.000311
.............................................................
Iteration 84     2022-05-01 11:52:54
   Deviance=173627.537 | Deviance change=0.004932
    Maximum b parameter change=0.000301
.............................................................
Iteration 85     2022-05-01 11:52:54
   Deviance=173627.5324 | Deviance change=0.004615
    Maximum b parameter change=0.000291
.............................................................
Iteration 86     2022-05-01 11:52:54
   Deviance=173627.5281 | Deviance change=0.004319
    Maximum b parameter change=0.000282
.............................................................
Iteration 87     2022-05-01 11:52:55
   Deviance=173627.524 | Deviance change=0.004042
    Maximum b parameter change=0.000272
.............................................................
Iteration 88     2022-05-01 11:52:55
   Deviance=173627.5202 | Deviance change=0.003783
    Maximum b parameter change=0.000263
.............................................................
Iteration 89     2022-05-01 11:52:55
   Deviance=173627.5167 | Deviance change=0.00354
    Maximum b parameter change=0.000255
.............................................................
Iteration 90     2022-05-01 11:52:55
   Deviance=173627.5134 | Deviance change=0.003313
    Maximum b parameter change=0.000247
.............................................................
Iteration 91     2022-05-01 11:52:55
   Deviance=173627.5103 | Deviance change=0.0031
    Maximum b parameter change=0.000239
.............................................................
Iteration 92     2022-05-01 11:52:55
   Deviance=173627.5074 | Deviance change=0.002901
    Maximum b parameter change=0.000231
.............................................................
Iteration 93     2022-05-01 11:52:55
   Deviance=173627.5047 | Deviance change=0.002715
    Maximum b parameter change=0.000223
.............................................................
Iteration 94     2022-05-01 11:52:55
   Deviance=173627.5021 | Deviance change=0.002541
    Maximum b parameter change=0.000216
.............................................................
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Iteration 95     2022-05-01 11:52:55
   Deviance=173627.4997 | Deviance change=0.002378
    Maximum b parameter change=0.000209
.............................................................
Iteration 96     2022-05-01 11:52:55
   Deviance=173627.4975 | Deviance change=0.002225
    Maximum b parameter change=0.000202
.............................................................
Iteration 97     2022-05-01 11:52:55
   Deviance=173627.4954 | Deviance change=0.002083
    Maximum b parameter change=0.000196
.............................................................
Iteration 98     2022-05-01 11:52:55
   Deviance=173627.4935 | Deviance change=0.001949
    Maximum b parameter change=0.000189
.............................................................
Iteration 99     2022-05-01 11:52:55
   Deviance=173627.4917 | Deviance change=0.001824
    Maximum b parameter change=0.000183
.............................................................
Iteration 100     2022-05-01 11:52:55
   Deviance=173627.49 | Deviance change=0.001707
    Maximum b parameter change=0.000177
.............................................................
Iteration 101     2022-05-01 11:52:55
   Deviance=173627.4884 | Deviance change=0.001598
    Maximum b parameter change=0.000171
.............................................................
Iteration 102     2022-05-01 11:52:55
   Deviance=173627.4869 | Deviance change=0.001495
    Maximum b parameter change=0.000166
.............................................................
Iteration 103     2022-05-01 11:52:55
   Deviance=173627.4855 | Deviance change=0.001399
    Maximum b parameter change=0.00016
.............................................................
Iteration 104     2022-05-01 11:52:55
   Deviance=173627.4842 | Deviance change=0.001309
    Maximum b parameter change=0.000155
.............................................................
Iteration 105     2022-05-01 11:52:55
   Deviance=173627.4829 | Deviance change=0.001225
    Maximum b parameter change=0.00015
.............................................................
Iteration 106     2022-05-01 11:52:55
   Deviance=173627.4818 | Deviance change=0.001147
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    Maximum b parameter change=0.000145
.............................................................
Iteration 107     2022-05-01 11:52:55
   Deviance=173627.4807 | Deviance change=0.001073
    Maximum b parameter change=0.000141
.............................................................
Iteration 108     2022-05-01 11:52:55
   Deviance=173627.4797 | Deviance change=0.001004
    Maximum b parameter change=0.000136
.............................................................
Iteration 109     2022-05-01 11:52:55
   Deviance=173627.4788 | Deviance change=0.00094
    Maximum b parameter change=0.000132
.............................................................
Iteration 110     2022-05-01 11:52:55
   Deviance=173627.4779 | Deviance change=0.00088
    Maximum b parameter change=0.000127
.............................................................
Iteration 111     2022-05-01 11:52:55
   Deviance=173627.4771 | Deviance change=0.000823
    Maximum b parameter change=0.000123
.............................................................
Iteration 112     2022-05-01 11:52:55
   Deviance=173627.4763 | Deviance change=0.00077
    Maximum b parameter change=0.000119
.............................................................
Iteration 113     2022-05-01 11:52:56
   Deviance=173627.4756 | Deviance change=0.000721
    Maximum b parameter change=0.000115
.............................................................
Iteration 114     2022-05-01 11:52:56
   Deviance=173627.4749 | Deviance change=0.000675
    Maximum b parameter change=0.000112
.............................................................
Iteration 115     2022-05-01 11:52:56
   Deviance=173627.4743 | Deviance change=0.000631
    Maximum b parameter change=0.000108
.............................................................
Iteration 116     2022-05-01 11:52:56
   Deviance=173627.4737 | Deviance change=0.000591
    Maximum b parameter change=0.000104
.............................................................
Iteration 117     2022-05-01 11:52:56
   Deviance=173627.4731 | Deviance change=0.000553
    Maximum b parameter change=0.000101
.............................................................
```

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Iteration 118     2022-05-01 11:52:56
   Deviance=173627.4726 | Deviance change=0.000517
    Maximum b parameter change=9.8e-05
------------------------------------------------------------
Start: 2022-05-01 11:52:50
End: 2022-05-01 11:52:56
Time difference of 5.74758 secs
Difference: 5.74758
------------------------------------------------------------

WLE Reliability= 0.953

Yen's Q3 Statistic based on an estimated theta score
*** 66 Items | 2145 item pairs
*** Q3 Descriptives
     M     SD    Min    10%    25%    50%    75%    90%    Max
-0.015  0.063 -0.148 -0.076 -0.052 -0.023  0.010  0.047  0.649
```

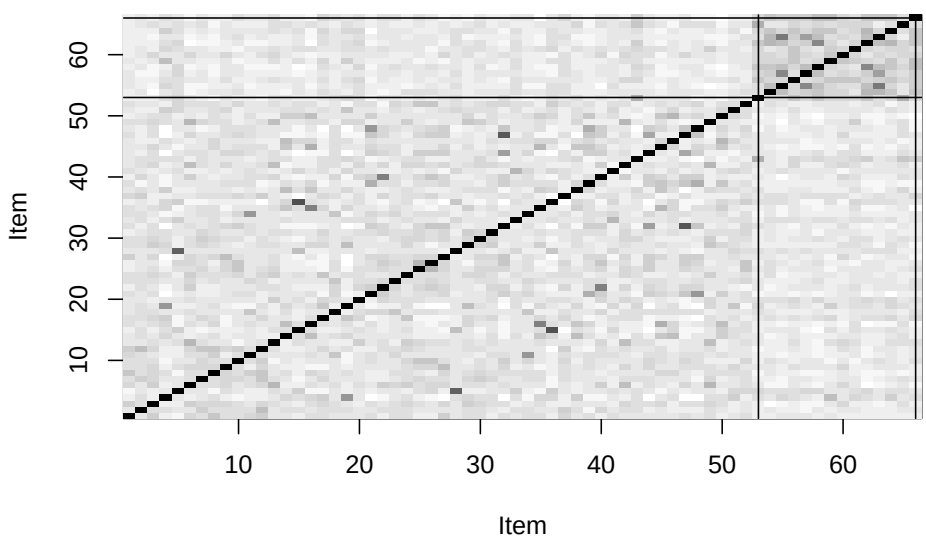


Figure 6.2: Correlations matrix of Item Residuals

> (Some sample language) "Using these criteria, we found XX items in possession of local dependence. The items that had the highest raw residual correlations had negative signs and no positive residual correlation. In other words, the assumption of local independence in this study (was)/(was not) met."

Figure 6.2 is a plot of the matrix of correlations between item residuals.

Summary

These analyses offer mixed evidence of assumption satisfaction. The model

explains 44% of the variance of the item responses. But we see three contrasts of the standardized residuals above the threshold of 2. In the Q3 plot we see a few strong correlations of item correlations.

6.1.2 Rating Scale Model (RSM)

The first IRT model is the Rating Scale Model (RSM) (citation). The model is an extension of the Rasch model (designed for dichotomous items) for use with polytomous items based on rating scales such as Likert scales of agreement [Embretson and Reise, 2000] and estimates the probability of choosing from among several response options given an examinee's location on the latent trait (level of lifetime exposure to spiritual abuse). More specifically, the increase in difficulty involved in choosing from between two response categories is called a "step parameter". The RSM assumes that all items use the same rating scale. As a Rasch model, it assumes that all items discriminate equally well and thus fixes the discrimination parameter to 1. The model also fixes the step parameters of all the items on the premise that the response categories are the same for all items. What the RSM frees to vary are the item difficulty parameters. Thus the focus of the RSM is the differences in item difficulties. If all of these assumptions are true – that the items and their rating scales differentiate examinees equally well and differ only in overall difficulty – then the observed pattern of item responses will closely match the pattern of items responses expected by the RSM and the model will fit.

Table 6.1: Rating Scale Model Item Parameters for Spiritual Harm and Abuse Items

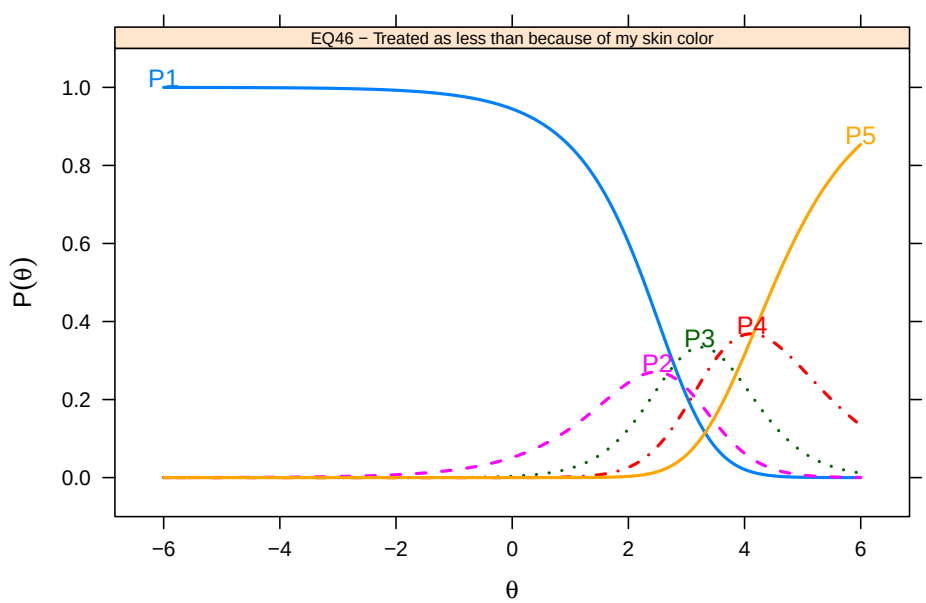
	a1	b1
EQ46 - Treated as less than because of my skin color	1	0.13
EQ41 - Denied opportunities to serve because of my sexual orientation	1	0.13
EQ17 - Treated as less than b/c my sexual orientation	1	0.13
EQ47 - Medical care being postponed/withheld for religious reasons	1	0.13
EQ60 - Encouraged by leader to stay in abusive marriage	1	0.13
EQ55 - Hearing cultural references in sermons unfamiliar to my race/eth. subculture	1	0.13
IQ72 - Feeling as if God harmed me directly	1	0.13
IQ81 - Nightmares about my negative religious experiences	1	0.13
EQ10 - Asked to give up personal goals by pastor	1	0.13
EQ65 - Cut off/shunned by more religious family members	1	0.13
EQ26 - Pressured to forgive abuser while abuse was ongoing	1	0.13
EQ24 - Expected to consult pastor/leader before making non-religious decisions	1	0.13
EQ23 - Prayer replacing needed medical interventions	1	0.13
EQ27 - Denied opportunities to serve b/c of my gender	1	0.13
EQ31 - Shunned/ignored by pastor/church/group	1	0.13
EQ30 - Deterred from seeking mental health treatment/counseling/medication	1	0.13
EQ43 - Shamed by pastor/group for poor spiritual/moral performance	1	0.13
IQ68 - Anxiety attacks triggered by religious stimuli	1	0.13

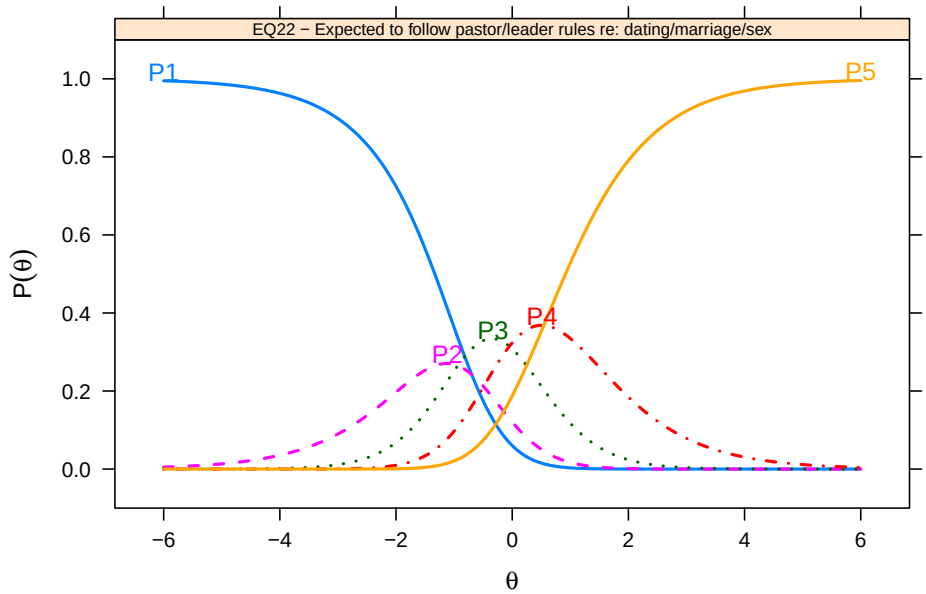
	a1	b1
EQ52 - Blamed for harm I suffered rather than blaming who harmed me	1	0.13
IQ70 - Feeling betrayed by God	1	0.13
EQ49 - Leadership/group protecting and elevating abusive individuals	1	0.13
EQ16 - Church/community abandoning me in difficult time	1	0.13
EQ20 - Members pressured to give money despite financial hardship	1	0.13
EQ57 - Developing mental/physical ailments from conforming to group/leader's expectation	1	0.13
EQ62 - Scripture used to justify physical violence	1	0.13
EQ59 - Pastor/group blame victim for their abuse	1	0.13
EQ21 - Taught I would risk Hell if left my church/group	1	0.13
IQ82 - Having trouble navigating life outside my church/community	1	0.13
EQ50 - Treated as less than because of my gender	1	0.13
EQ64 - Witnessing women pressured to stay in unfaithful/abusive marriages	1	0.13
EQ28 - Leadershi/group protecting abusive individuals	1	0.13
EQ40 - Threatening Divine punishment to keep group members in line	1	0.13
EQ63 - Shamed by pastor/group for raising questions or concerns	1	0.13
EQ19 - Behavior excessively monitored by pastor/group	1	0.13
EQ51 - Extreme pressure to be pastor/missionary/spiritual leader	1	0.13
EQ53 - Scripture used to justify abusive parent-child behavior	1	0.13
IQ79 - Distrust of God	1	0.13
EQ34 - Scripture used to justify physical punishment/severe discipline	1	0.13
EQ29 - Feeling special when in pastor's good graces; otherwise ignored	1	0.13
IQ73 - Self-hatred or self-loathing	1	0.13
EQ11 - Disagree w/pastor portrayed as evil	1	0.13
EQ7 - Pastor speaking on God's behalf	1	0.13
EQ58 - Terror/horror used to motivate religious decisions	1	0.13
EQ42 - Seeing others shamed/shunned by pastor/leader/group	1	0.13
EQ44 - Love/acceptance offered only for high spiritual/moral performance	1	0.13
EQ56 - Being made to feel I was crazy/weird for doubts/questions	1	0.13
EQ18 - Church/pastor discourage critical thinking	1	0.13
IQ78 - A lack of self-worth	1	0.13
EQ45 - Developmentally-inappropriate/anxious Hell/Satan/demons taught to young children	1	0.13
IQ74 - Sadness over the loss of my faith/religious community	1	0.13
EQ36 - Mental/physical problems interpreted as spiritual/moral weakness	1	0.13
EQ61 - Developmentally-inappropriate/anxious end times descriptions taught to young children	1	0.13
EQ32 - Feeling unable to express unhappiness	1	0.13
IQ76 - Feeling I wasted years of my life in a church/set of beliefs	1	0.13
EQ54 - Being explicitly taught to distrust my intuitions	1	0.13
EQ37 - Made to feel less spiritually mature than pastor/leadership	1	0.13
EQ38 - Unrealistic demands placed on my moral/religious behavior	1	0.13
IQ71 - Avoiding religious activities/settings to reduce distressing feelings	1	0.13
IQ69 - Lack of spiritual direction/purpose	1	0.13
IQ80 - Feeling isolated	1	0.13
EQ39 - Made to feel shame over natural sexual desires (not actions)	1	0.13
IQ77 - Anger upon reflecting on negative religious experiences	1	0.13

	a1	b1
EQ35 - Taught to distrust my emotions	1	0.13
EQ25 - Feeling unable to raise questions and issues	1	0.13
EQ66 - Others treated as less than due to their sexual orientation	1	0.13
EQ22 - Expected to follow pastor/leader rules re: dating/marriage/sex	1	0.13

Item parameters. Table 6.1 reports the item parameters estimated by the RSM for all 66 SHAS items. The first column shows the discrimination parameter, fixed to "1" for all items on the premise that the items discriminate equally well. The subsequent columns (b1 to b4) report the estimated thresholds (or step parameters). These values, too, are the same for all items.

Item plots offer a way to see this information visually. Following are plots of three items. Here the horizontal (x) axis is the scale of the latent trait (overall lifetime exposure to spiritual abuse) expressed in logits ranging from -6 to +6. The vertical (y) axis is the probability of selecting a given response (such as "Always") at each level of latent trait. The blue line is the probability of selecting "Never" for examinees ranging from low to high lifetime spiritual abuse.





What does vary between items in the RSM is the difficulty parameter, shown in the "c" column of Table 6.1. This represents the location of the item on the logit scale of the latent trait: Does the item represent the "most" spiritual abuse? The least? Or somewhere in the middle? Table 6.1 presents the items sorted from least abuse to most abuse.

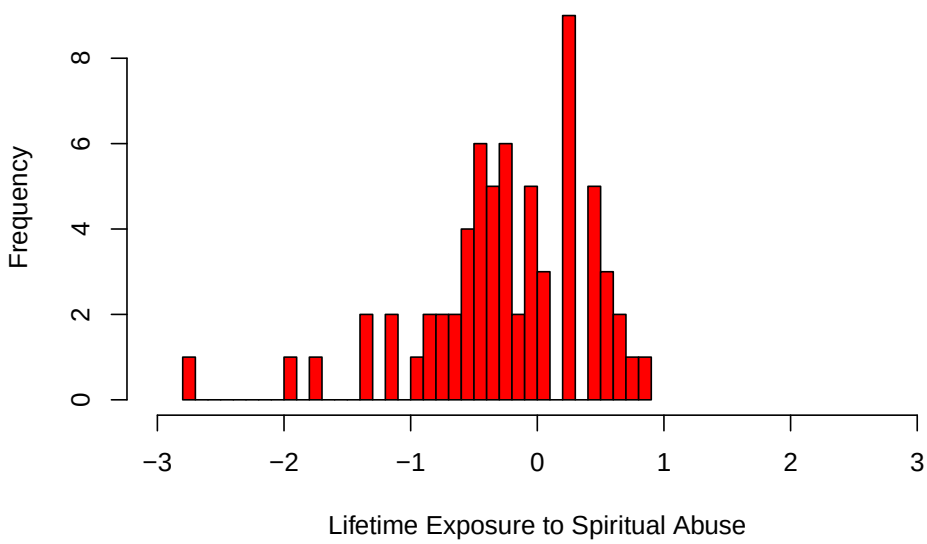
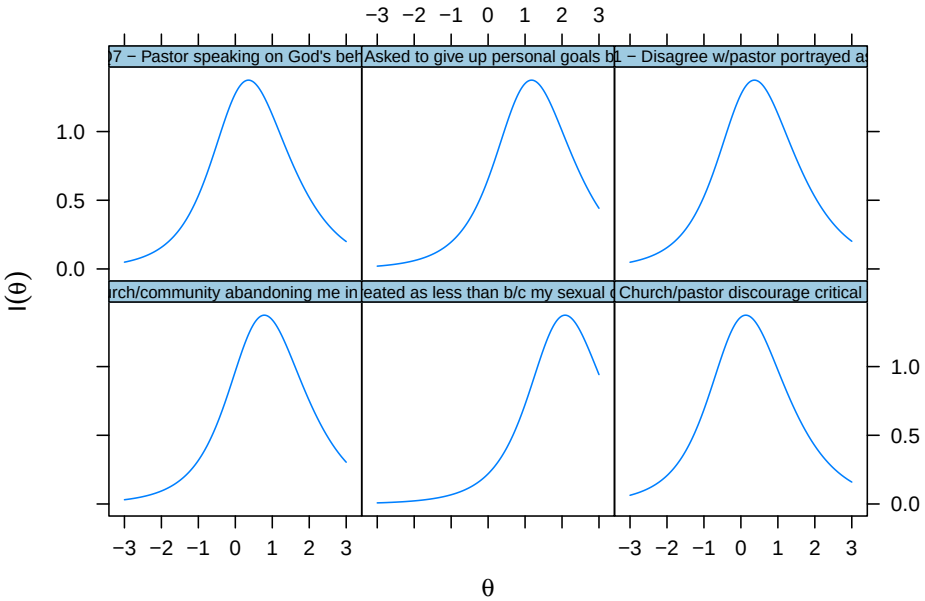


Figure 6.3: Distribution of Item Difficulty

Figure 6.3 presents a histogram of the distribution of item difficulty parameters.

The distribution is negatively skewed with most items clustered slightly above the average level of spiritual and several items measuring very low levels of lifetime exposure to spiritual abuse.



Item fit statistics. Fit statistics show how closely the observed pattern of item responses fits the pattern of item responses predicted by the particular IRT model (Rasch, partial credit, etc.). They do this by means of chi-square statistics which examine the cumulative difference the observed pattern of item responses and the pattern of item responses that the model would expect. Two fit statistics commonly used in IRT models are the infit mean square (MNSQ) and the outfit mean square (MNSQ) [Bond and Fox, 2015]. The infit statistic places greater emphasis on unexpected responses that are close to the examinees and item location. The outfit is sensitive to unexpected responses that are far from the location. The expected value of infit or outfit for each item is 1.0, with a range of acceptable values ranging from 0.5 to 1.5. Values outside these boundaries indicate a lack of fit between items and the model.

Table 6.2: Rating Scale Model (RSM) - Misfitting Items

item	outfit	z.outfit	infit
EQ17 - Treated as less than b/c my sexual orientation	1.89	10.09	2.26
EQ27 - Denied opportunities to serve b/c of my gender	2.02	18.55	1.77
EQ41 - Denied opportunities to serve because of my sexual orientation	2.63	14.93	2.26
EQ46 - Treated as less than because of my skin color	2.37	8.95	1.89
EQ50 - Treated as less than because of my gender	1.68	15.51	1.73
EQ55 - Hearing cultural references in sermons unfamiliar to my race/eth. subculture	2.68	21.37	1.62

All but six of the 66 items had infit and outfit statistics within the acceptable range from 0.5 to 1.5. Statistics for these six misfitting items are presented here in Table 6.2. The large outfit values mean the observed patterns of responses to these items differed substantially from the patterns of responses expected by the PCM. These items capture the experiences of women, people of color, and LGBTQ persons.

Model fit. To assess the overall fit of the model to the item response data we use two statistics. The first is the Root Mean Square Error of Approximation (RMSEA). The suggested cutoff for RMSEA is .06. The second is the CFI. The suggested cutoff for the CFI is .95.

Table 6.3: Rating Scale Model Fit - Overall Model Fit

	M2	df	p	RMSEA	RMSEA_5	RMSEA_95	TLI	CFI
stats	61561.13	2339	0	0.09	0.09	0.09	0.96	0.95

Table 6.3 reports the model fit statistics for the SHAS items. The obtained RMSEA value = .092 (95% CI[.092, .093]) exceeds the cutoff of .06. The CFI = .953 meets the recommended .95 threshold. Together these statistics offer weak evidence that the RSM fits the SHAS items.

6.1.3 Partial Credit Model (PCM)

The second model we consider is the Partial Credit Model (PCM) [Masters, 1982]. This is a polytomous item response model belonging to the Rasch family. This means that it extends the Rasch model, which is based on dichotomous item responses (1=correct, 0=incorrect), by considering the probability of choosing from among several response options such as those typically presented in Likert scales of agreement. The number of response categories in each item may vary [Embretson and Reise, 2000]. It follows that the PCM contains m (m + 1 being the number of response categories) threshold parameters, and each threshold parameter marks a category intersection [Masters, 1982]. Threshold sometimes refers to "step difficulty" or "step parameter" and illustrates the point on the latent trait continuum (e.g., spiritual abuse) where a response in category k becomes more likely than a response in category k − 1 (de Ayala, 2009). The PCM provides scores for items, persons, and step parameters on a logit scale. We begin by loading the `mirt` library and estimating the model on the data from the 66 SHAS items. This model constrains the slopes to 1 and freely estimates the variance parameters.

Table 6.4: Partial Credit Model - Item Parameters - I
Category Threshold Reversals

	a	b1
EQ41 - Denied opportunities to serve because of my sexual orientation	1	3.87
EQ17 - Treated as less than b/c my sexual orientation	1	3.94
EQ46 - Treated as less than because of my skin color	1	3.89
EQ27 - Denied opportunities to serve b/c of my gender	1	1.62
EQ50 - Treated as less than because of my gender	1	1.57
EQ21 - Taught I would risk Hell if left my church/group	1	1.08
EQ60 - Encouraged by leader to stay in abusive marriage	1	2.21
EQ26 - Pressured to forgive abuser while abuse was ongoing	1	1.40
EQ20 - Members pressured to give money despite financial hardship	1	0.96
EQ10 - Asked to give up personal goals by pastor	1	1.25
EQ40 - Threatening Divine punishment to keep group members in line	1	0.64
EQ65 - Cut off/shunned by more religious family members	1	1.19
EQ57 - Developing mental/physical ailments from conforming to group/leader's expectation	1	0.81
EQ47 - Medical care being postponed/withheld for religious reasons	1	1.55
EQ22 - Expected to follow pastor/leader rules re: dating/marriage/sex	1	-0.41
IQ76 - Feeling I wasted years of my life in a church/set of beliefs	1	-0.04
IQ72 - Feeling as if God harmed me directly	1	1.35
EQ39 - Made to feel shame over natural sexual desires (not actions)	1	-0.19
EQ24 - Expected to consult pastor/leader before making non-religious decisions	1	1.10
EQ7 - Pastor speaking on God's behalf	1	0.20
EQ61 - Developmentally-inappropriate/anxious end times descriptions taught to young children	1	-0.02
EQ51 - Extreme pressure to be pastor/missionary/spiritual leader	1	0.41
EQ30 - Deterred from seeking mental health treatment/counseling/medication	1	0.85
EQ53 - Scripture used to justify abusive parent-child behavior	1	0.33
EQ55 - Hearing cultural references in sermons unfamiliar to my race/eth. subculture	1	1.32
IQ68 - Anxiety attacks triggered by religious stimuli	1	0.83
EQ52 - Blamed for harm I suffered rather than blaming who harmed me	1	0.69
EQ19 - Behavior excessively monitored by pastor/group	1	0.41
EQ66 - Others treated as less than due to their sexual orientation	1	-0.62
IQ81 - Nightmares about my negative religious experiences	1	0.91
EQ54 - Being explicitly taught to distrust my intuitions	1	-0.39
EQ62 - Scripture used to justify physical violence	1	0.52
EQ23 - Prayer replacing needed medical interventions	1	0.74
EQ58 - Terror/horror used to motivate religious decisions	1	0.01
EQ38 - Unrealistic demands placed on my moral/religious behavior	1	-0.49
EQ29 - Feeling special when in pastor's good graces; otherwise ignored	1	0.09
EQ34 - Scripture used to justify physical punishment/severe discipline	1	0.07
EQ35 - Taught to distrust my emotions	1	-0.65
EQ18 - Church/pastor discourage critical thinking	1	-0.19
EQ44 - Love/acceptance offered only for high spiritual/moral performance	1	-0.03
EQ56 - Being made to feel I was crazy/weird for doubts/questions	1	-0.22

	a	b1
EQ32 - Feeling unable to express unhappiness	1	-0.35

Item parameters. Table 6.4 reports the item parameters estimated by the PCM. Once again, column (a) reports the discrimination parameter fixed as "1" for all items on the premise that the items discriminate examinees equally well. The subsequent columns (b1 to b4) report the estimated thresholds (or step parameters) for each item. Because the SHAS items all have five response categories, the PCM estimates four threshold parameters (b1 to b4) for each item. These represent the change in lifetime exposure to spiritual abuse represented by choosing between two response categories, such as, for example, "Once or twice" over "Never." Unlike the Rating Scale Model (RSM) which estimated the same thresholds for all items and allowed the items to vary only by overall difficulty, the PCM allows the thresholds to vary [Embretson and Reise, 2000]. The important analytic question, therefore, is which items have ordered thresholds and which have "reversals." A reversal between two category thresholds suggests that one might not be contributing reliable information and might be collapsed. Most of the SHAS items (43 out of 66) had reversals, and these items with reversals are shown in Table 6.4. On each of these items, examinees who had experienced more overall lifetime spiritual abuse were more likely to report "Never" than "Once or twice".

Again, plots of the option characteristic curves help convey this information visually. The following plots illustrate three items with very different sets of curves. The first item illustrates the intended pattern of ordered thresholds. The second and third items show pronounced reversals.

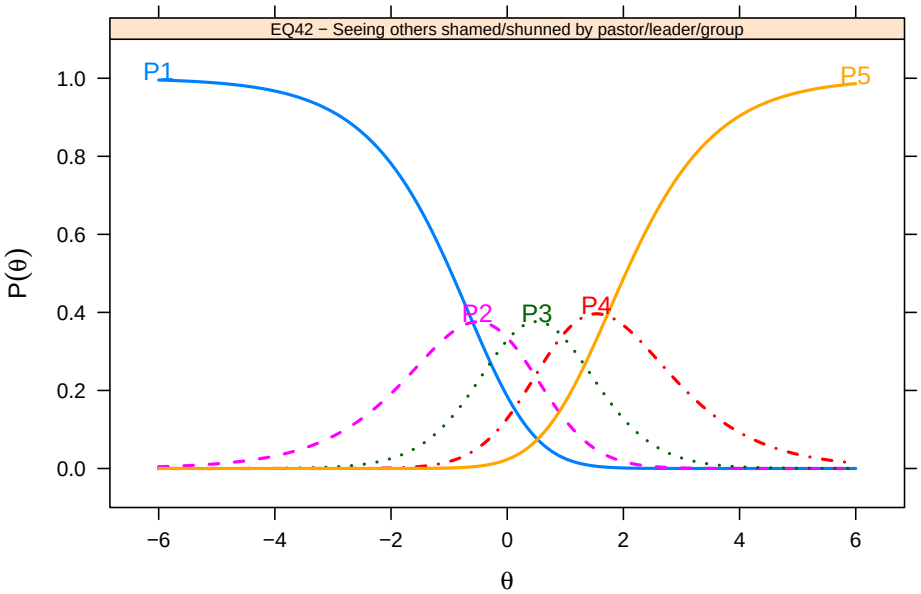


Figure **??** shows the intended pattern of responses. Examinees who had experienced less lifetime exposure to spiritual abuse were most likely to select P1: "Never." Those who had experienced a little more were more likely to select P2: "Once or twice." And so on. On this item the response categories functioned as intended to sort examinees on the basis of lifetime exposure to spiritual abuse.

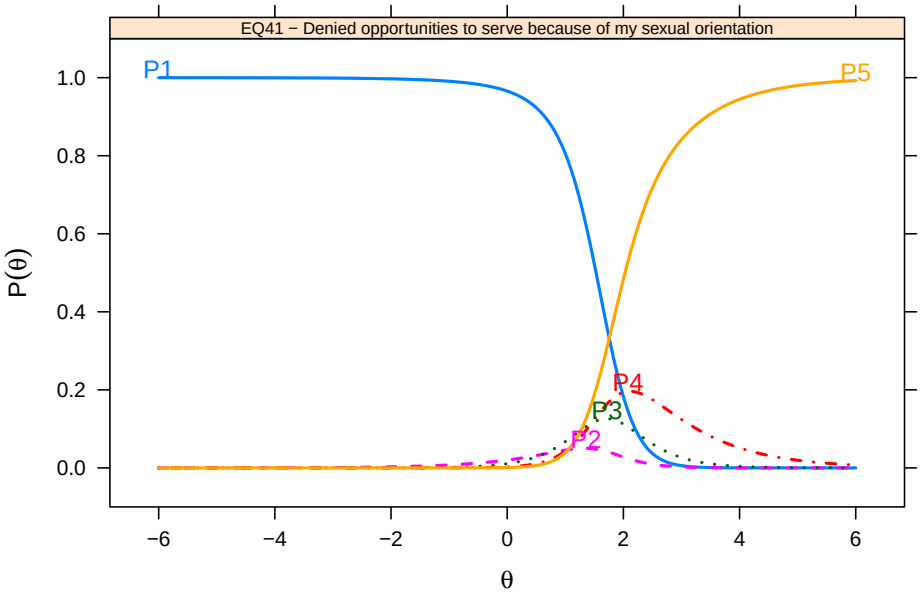
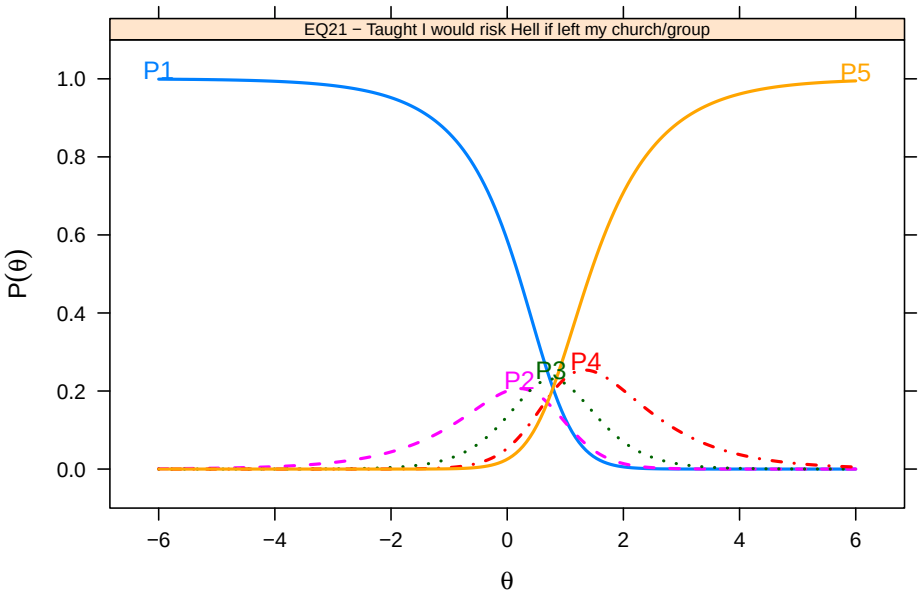


Figure **??** shows an item with reversals. the intended pattern of responses. Examinees who had experienced less lifetime exposure to spiritual abuse were most likely to select P1: "Never." Those who had experienced a little more were more likely to select P2: "Once or twice." And so on. On this item the response categories functioned as intended to sort examinees on the basis of lifetime exposure to spiritual abuse.



Fit statistics. Fit statistics gauge the extent to which the observed pattern of item responses fit the IRT model (Rasch, partial credit, etc.). They do this by means of chi-square statistics which examine the cumulative difference the observed pattern of item responses and the pattern of item respones that the model would expect. Two fit statistics commonly used in IRT modesl are the infit mean square (MNSQ) and the outfit mean square (MNSQ) [Bond and Fox, 2015]. The infit statistic places greater emphasis on unexpected responses that are close to the examinees and item location. The outfit is sensitive to unexpected responses that are far from the location. The expected value of infit or outfit for each item is 1.0, with a range of acceptable values ranging from 0.5 to 1.5. Values outside these boundaries indicate a lack of fit between items and the model.

Table 6.5: Partial Credit Model (PCM) - Misfitting Items

item	outfit	z.outfit	infit
EQ17 - Treated as less than b/c my sexual orientation	2.79	6.96	1.55
EQ27 - Denied opportunities to serve b/c of my gender	2.68	17.94	1.50
EQ41 - Denied opportunities to serve because of my sexual orientation	6.62	14.95	1.52
EQ46 - Treated as less than because of my skin color	3.25	8.33	1.42
EQ50 - Treated as less than because of my gender	1.95	12.41	1.51
EQ55 - Hearing cultural references in sermons unfamiliar to my race/eth. subculture	2.82	22.06	1.58

This information suggests a decent fit of the PCM to the SHAS item response data. All but six of the 66 items had infit and outfit statistics within the

acceptable range from 0.5 to 1.5. Statistics for these six misfitting items are presented here in Table 6.5. The large outfit values mean the observed patterns of responses to these items differed substantially from the patterns of responses expected by the PCM. These items capture the experiences of women, people of color, and LGBTQ persons.

Model fit.

```
            M2   df p      RMSEA    RMSEA_5   RMSEA_95     SRMSR     TLI
stats 51392.82 2144 0 0.08782602 0.08715423 0.08847002 0.1170549 0.96122
            CFI
stats 0.9612381
```

The obtained RMSEA value = .088 (95% CI[.088, .089]) and SRMSR value = .118 suggest that data do not fit the model reasonably very well using suggested cutoff values of RMSEA <= .06 and SRMSR <= .08 as suggested guidelines for assessing fit. The CFI = .961 exceeded the recommended .95 threshold.

Chapter 7

Reliability

In this chapter, we addresses the reliability of the SHAS based on the 66 piloted items. We begin by defining reliability and discussing its importance. Then we report the results of our analyses of the reliability of the piloted SHAS items from both classical test theory (CTT) and item response theory (IRT) frameworks. Since these are cross-sectional data taken at one point in time, our work is limited to *internal consistency* reliability which focuses on the relationships between the items at one point in time rather than relationships between items or scores between two or more time points. We conclude by considering what this information contributes to improved SHAS instrumentation.

7.1 About reliability

Reliability is the extent to which items and their scores are consistently measuring the trait of interest. It is evidence that two examinees who provide the same responses to the same items, and by extension receive the same total scores, truly have the same level or amount of, or exposure to, the trait.

7.2 Reliability in Classical Test Theory (CTT)

One way to think about reliability in classical test theory is the ratio of signal to noise in the observed data. Do the various item responses and scores provide a true sort of examinees on the trait, or is all this variation merely noise?

The foundation of internal consistency reliability in CTT is relationships between item responses: correlations. Strong correlations among the items are evidence that the items are measuring the same construct. Thus we begin by examining correlation matrices.

Figure 7.1 is a correlation matrix of the External Events items for the *3,174*

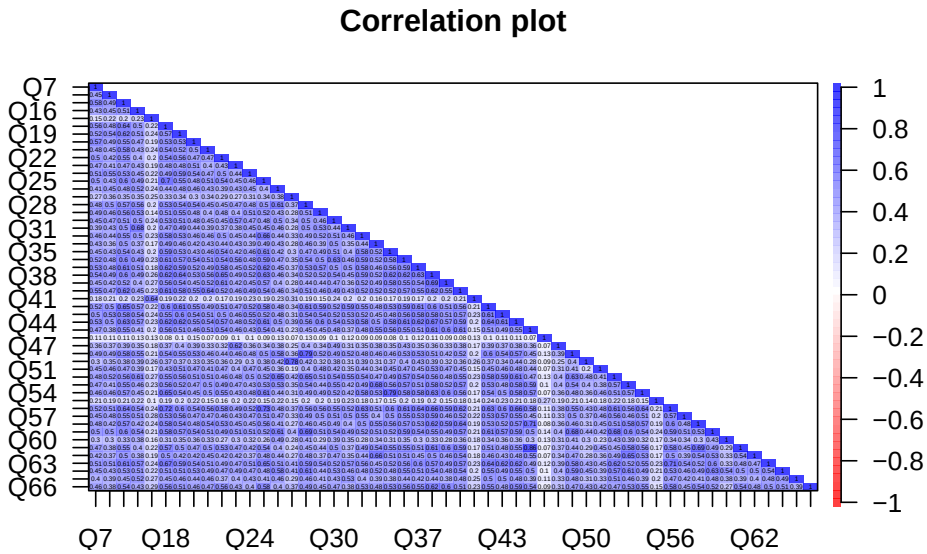


Figure 7.1: Correlation Matrix of External Events Items

examinees who responded to all of these items. Each cell reports the correlation coefficient between scores of the two items. All of the cells are color coded to the scale of correlations which range from 0 (indicating no relationship) to 1 (indicating a perfect linear relationship). The cells along the diagonal are all 1.0 because they represent the correlation of the item with itself. Only one half of the matrix is necessary to report all the correlations among all the items.

The plot offers several interesting pieces of information. The first is that Q17–"Being treated as 'less than' because of my sexual orientation–does not quite fit. Responses to this item correlate weakly with responses to all the other items. It is possible that so few examinees experience this event that the responses vary very little, which would attenuate any correlation. Or it may be measuring something different than what the other items in this set. Similar could be said of Q27–"Being denied opportunities to serve/volunteer because of my gender."

Figure 7.2 is a correlation matrix of the Internal States items for the *3,051* examinees who responded to all of these items. This plot is interesting in that most of the correlations are in the moderate (0.50) range. They are sorting examinees in a similar way, but together they do not measure exactly the same thing.

This final matrix, Figure 7.3, reports the correlations among all of the Likert items among the *2,979* examinees who responded to all 66 items. Note that missing data are a challenge here, as these plots require complete data.

This plot offers several interesting insights. First, the preponderance of blue indicates that everything is positively correlated. This makes some sense because

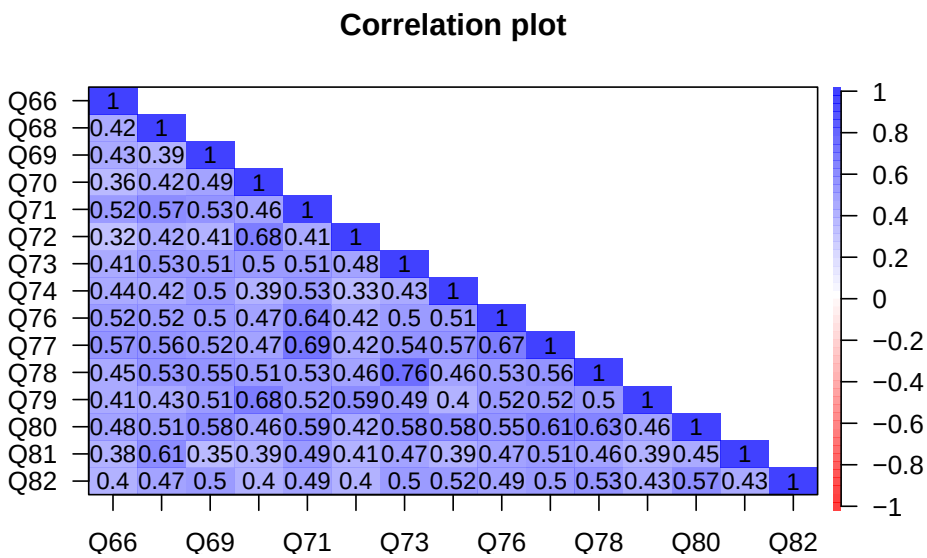


Figure 7.2: Correlation Matrix of Internal States Items

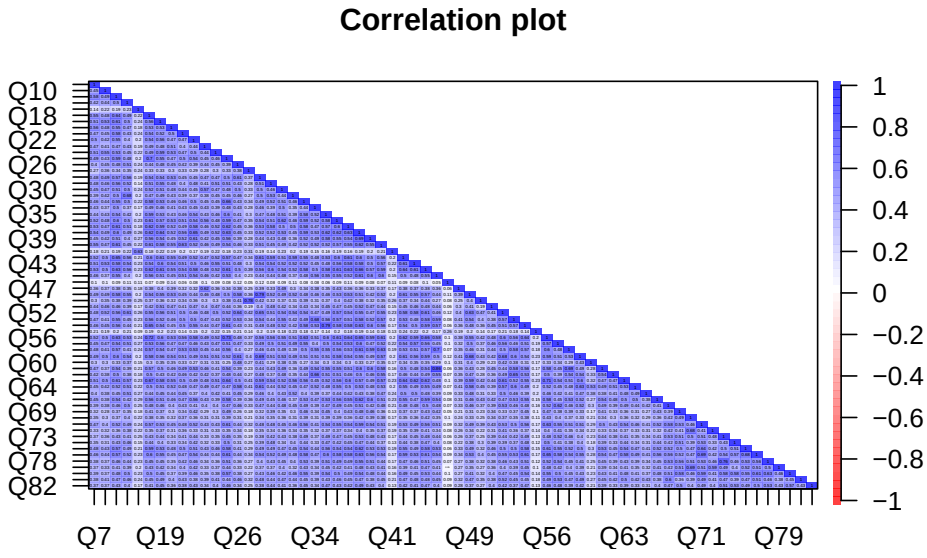


Figure 7.3: Correlation Matrix of All 66 Likert SHAS Items

the items were worded and all the response categories were coded in the same direction.

Second, in this plot there are too many items and correlation coefficients in one place to be distinguishable, but with the colors it is possible to see an overall pattern. This plot orders the items by their order on the questionnaire so we can see how the External Events items correlate with the Internal Events items: moderately. It is possible that these are not two different scales.

Finally, we see at a glance a handful of items that correlate very weakly with all the other items. These items should be scrutinized for refinement and possible omission from the analysis.

Cronbach's coefficient alpha. Arguably the most well-known statistic for estimating internal consistency reliability is Cronbach's coefficient alpha (citation). This is a measure of the average inter-item correlation (?). It is also a measure of the ratio of signal to noise (X = T + e). A ballpark cutoff value is 0.70 [Nunnally and Bernstein, 1994].

```
Number of Items
66

Number of Examinees
2979

Coefficient Alpha
0.981
```

This output is not pretty, but it does report a Cronbach's alpha of 0.98. This is a high level of internal consistency reliability.

7.3 Reliability in Item Response Theory (IRT)

Wright and Masters (1982) developed reliability measures based on the Rasch measurement model, which have a different concept from classical test theory (e.g., Cronbach's alpha). Reliability is estimated both for persons and items, by means of Person Separation Reliability (PSR). This is an estimate of how well this instrument can distinguish respondents on the measured variable. In parallel, Item Separation Reliability (ISR) is an indication of how well items are separated by the persons taking the test (Wright & Stone, 1999). The cutoff of separation reliabilities is > 0.80 (Bond & Fox, 2015).

7.4 Summary

Chapter 8

Scale Analysis

In this chapter, we examine the technical quality of the scales derived from the SHAS piloted items. Here we address questions like the following:

- **Person fit** - How well does the current scale fit the population of examinees? Are there ceiling or floor effects? To what extent might the existing items under-represent the full range of abuse?
- **Scale invariance** - Do the scales mean the same thing to different groups of examinees? To what extent do these scales (abuse constructs) vary by age, gender, and/or sexuality?
- **Developmental pathway** - Beyond total scores, what might an incremental scale of in terms of items look like? What items might represent the most severe or most frequent forms of spiritual abuse?

We begin by defining scaling and discussing the importance of scale analysis. We report the results of our scale analysis from within the item response theory (IRT) framework. We conclude by considering what this information contributes to improved SHAS instrumentation.

8.1 Person Fit

8.2 Scale Invariance

8.3 The Item-Person Map

The item-person map (also known as the Wright map) is a tool for visualizing the results of a calibration of items and persons. The Wright map was originally developed for use in test development but it also helps illuminate the results of surveys [Bulut, 2021]. As Bulut [2021] explains:

> "The Partial Credit Model (PCM), which is an extension of the Rasch model for polytomous item responses, can be used for survey development and validation (Green & Frantom, 2002). We can analyze survey items using PCM and obtain item thresholds for each item. These thresholds indicate the magnitude of the latent trait required to select a particular response option (e.g., selecting disagree over strongly disagree). Using an item-person map, we can see thresholds for the items as well as the distribution of the latent trait for the respondents. This plot is quite useful for analyzing the alignment (or match) between the items and respondents answering the items."

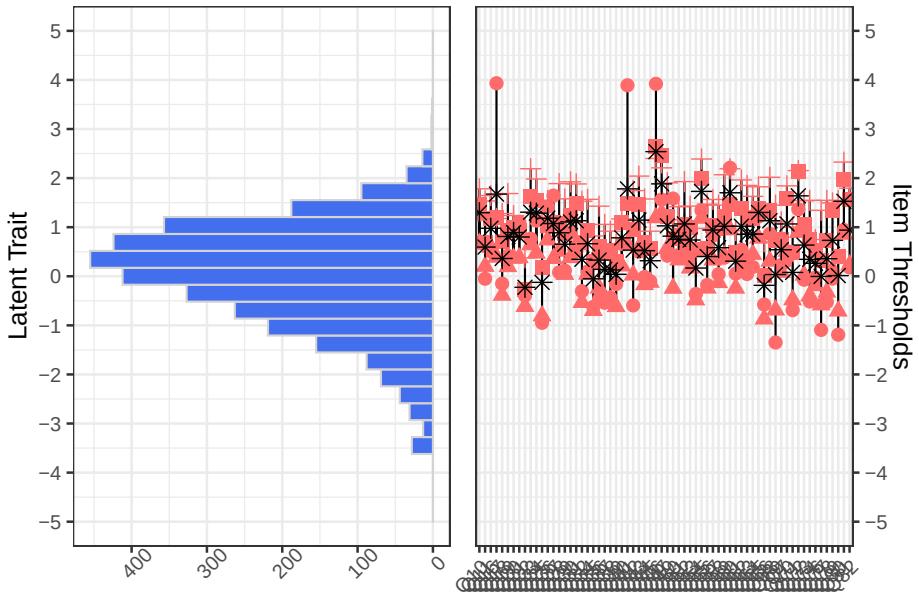


Figure 8.1: Item-Person Map: All Likert Items

Figure 8.1 is an item-person map of all 66 Likert SHAS items against all of the examinees. In the plot, red points indicate item thresholds for each item (three thresholds separating four response categories) and the asterisk indicates the average value of the thresholds for each item. The higher these thresholds, the more spiritual abuse the examinee has experienced. On the left-hand side of the plot, we also see the distribution of examinees along the latent trait (i.e., the construct of spiritual abuse).

This plot offers several insights at a general level. First, the plot of examinees on the left suggests that the latent trait of spiritual abuse is normally distributed, with a slight negative skew. Very few examinees reported very low or very high levels of exposure to spiritual abuse; most reported a moderate to slightly higher than average level of spiritual abuse. Second, with this plot it is difficult

to distinguish individual items but it does show that they tend to clusteraround this “slightly higher than average” level of spiritual abuse, with none capturing a very low level of abuse and just a handful of items measuring much higher than average levels of spiritual abuse. Three items had response categories that captured very high levels of spiritual abuse.

8.4 Information curves

These provide important information about where along the scale of the latent trait the item or set of items provides the most reliable information. If there are too few “easy” or “difficult” items, then these extreme scores for examinees are less reliable.

8.5 Person Separation Reliability

8.6 Item Separation

8.7 Summary

Chapter 9

Discussion

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